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Oefeningenreeks 2B nr. 6.4

$$\text{Bgtan}\,\frac{1}{2} + \text{Bgtan}\,\frac{1}{5} + \text{Bgtan}\,\frac{1}{8} = \frac{\pi}{4}$$

Stel:

$$\alpha = \text{Bgtan}\,\frac{1}{2} \implies \tan\alpha = \frac{1}{2}$$

$$\beta = \text{Bgtan}\,\frac{1}{5} \implies \tan\beta = \frac{1}{5}$$

$$\gamma = \text{Bgtan}\,\frac{1}{8} \implies \tan\gamma = \frac{1}{8}$$

$$\tan(\alpha+\beta) = \frac{\frac{1}{2}+\frac{1}{5}}{1-\frac{1}{2}\cdot\frac{1}{5}} = \frac{7}{9}$$

$$\tan(\alpha+\beta+\gamma) = \frac{\frac{7}{9}+\frac{1}{8}}{1-\frac{7}{9}\cdot\frac{1}{8}} = 1$$

$$\tan(\alpha+\beta+\gamma) = 1$$

$$\tan\left(\frac{\pi}{4}\right) = 1$$

$$\alpha+\beta+\gamma = 1$$

$$\text{Bgtan}(1) = \frac{\pi}{4}$$

References